68 questions | **~150** parts | **Exam-style** | **File** 7 / 8

Q Chapter 3 exam-style questions

أسئلة امتحانية حقيقية: تفسير، مسائل عكسية، مقارنة، كشف أخطاء. ليست أسئلة مباشرة

Built from the course outline's assessment criteria: **CLO2-A** (calculate central tendency, variability and position), **CLO3-B, C** (extract and reason from graphs and summary statistics) and **CLO1-A** (parameters vs statistics). Every question carries its lesson and outcome tags. If you miss a question, the lesson tag tells you which file to reopen. Answers are in a **separate file** (07); work closed-book first.

LESSON	3-1	3-2	3-3	3-4
Questions touching the lesson	22	28	17	11

طريقة الاستخدام

الأسئلة هنا **ليست مباشرة**، بل على نمط أسئلة الكويز والميدتيرم: قيمة مفقودة، تصحيح خطأ في التسجيل، مقارنة بين مجموعتين، قراءة مخطط صندوقي، اكتشاف خطأ طالب، وتبرير صح أو خطأ. حلّها **مغلق الكتاب**، ثم صحّح من ملف الإجابات (07). الجزء **F** محاكاة للكويز الصفي الأول: 30 دقيقة لـ 20 درجة.

Part A · Concept and interpretation (multiple choice)

Each item needs a reason, not a recalled fact. Give the letter *and* one line of justification, as in a written exam.

A1 3-1 CLO1-A

A hospital records the mean waiting time of **all** 1,240 patients who visited its emergency department in 2025: 47 minutes. The value 47 is:

(a) a statistic, because it is a mean

(b) a parameter, because it describes the entire population of interest

(c) a statistic, because it was computed from recorded data

(d) a parameter only if the data are normally distributed

A2 3-1 CLO3-C

In a company the mean monthly salary is OMR 1,450 and the median is OMR 820. Which conclusion is best justified?

(a) the salary distribution is symmetric

(b) a few very high salaries pull the mean up, so the median better represents a typical employee

(c) most employees earn more than OMR 1,450

(d) the mode must be greater than OMR 1,450

A3 3-1 CLO2-A

A frequency table of household incomes has a last class "OMR 3,000 and above". Which measure **cannot** be computed from the table?

(a) the modal class

(b) the median class

(c) the mean

(d) the total number of households

A4 3-1 CLO3-B

A distribution has mean 63, median 70 and mode 74. Its shape is:

(a) positively skewed

(b) negatively skewed

(c) symmetric

(d) impossible to tell without a graph

A5 3-1 CLO2-A

All 20 values in a data set equal 12. Which statement is **false**?

(a) mean = median = 12

(b) the mode is 12

(c) the standard deviation is 0

(d) the data set has no mode because no value differs from the others

A6 3-2 CLO3-C

Two machines fill 250 g packs. Both have mean 250 g. Machine A has s = 1.8 g and machine B has s = 4.1 g. Which is correct?

(a) A is more consistent

(b) B is more consistent

(c) they are equally consistent because the means are equal

(d) consistency cannot be compared without the CV

A7 3-2 CLO1-A

Which statement about the variance is **false**?

(a) it can equal zero

(b) it can be negative when most values are below the mean

(c) its units are the square of the data units

(d) dividing by n − 1 instead of n makes the sample variance slightly larger

A8 3-2 CLO2-A

Without knowing the shape of a distribution, the smallest percentage of values guaranteed within 2.5 standard deviations of the mean is:

(a) 68%

(b) 84%

(c) 95%

(d) 75%

A9 3-2 CLO2-A

Bell-shaped data have mean 60 and SD 5. Approximately what percentage of values are **above 70**?

(a) 5%
(b) 2.5%
(c) 16%
(d) 0.15%

A10 3-2 CLO3-C

Every employee in a company receives a raise of OMR 50 per month. What happens to the standard deviation and the coefficient of variation of salaries?

(a) both unchanged
(b) SD unchanged, CV decreases
(c) SD increases by 50, CV unchanged
(d) both increase

A11 3-2 CLO2-A

Every value in a data set with variance 6.5 is multiplied by 3. The new variance is:

(a) 6.5
(b) 19.5
(c) 58.5
(d) 9.5

A12 3-3 CLO3-C

Salma's z-score on a test is −0.8. This means:

(a) she answered 80% of the questions wrongly
(b) her score is 0.8 standard deviations below the class mean
(c) 80% of the class scored above her
(d) her score is 0.8 marks below the mean

A13 3-3 CLO3-C

Ali's score is at the 70th percentile of his class. Which statement is correct?

(a) Ali answered 70% of the questions correctly
(b) about 70% of the class scored lower than Ali
(c) Ali's score is 70
(d) Ali scored 70% above the mean

A14 3-3 CLO1-A

Which of these is **not** equal to the median?

(a) P_{50}
(b) Q_2
(c) D_5
(d) P_5

A15 3-3 CLO3-C

An extremely large value is added to a data set. Which measure changes the **most**?

(a) the median
(b) the IQR
(c) the standard deviation
(d) Q_1

A16 3-4 CLO3-B

In a boxplot, the median is close to Q_3 and the left whisker is three times as long as the right whisker. The distribution is:

(a) symmetric
(b) positively skewed
(c) negatively skewed
(d) bimodal

A17 3-4 CLO1-A

In exploratory data analysis, the measures of centre and of variation are respectively:

(a) mean and standard deviation
(b) median and IQR
(c) mode and range
(d) median and standard deviation

A18 3-4 CLO3-B

In a boxplot, the right whisker (Q_3 to max) is twice as long as the left half of the box (Q_1 to median). This tells you that:

(a) more values lie above Q_3 than between Q_1 and the median
(b) the top 25% of values are more spread out than the second quarter
(c) the data contain an outlier
(d) the mean is above Q_3

Part B · Short exam problems by lesson

Inverse problems, corrections, comparisons and interpretation, the way quiz and midterm questions are actually built.

B1 3-1 CLO2-A

The mean of six sample values is 14.5. Five of them are 12, 18, 9, 16 and 20. Find the sixth value, then the median and mode of all six.

B2 3-1 CLO2-A

The mean of 25 exam scores was reported as 68. It was later found that one score had been recorded as 48 instead of 84. (a) Find the correct mean. (b) Must the median change? Explain.

B3 3-1 CLO2-A · CLO3-C

Section 10 of STAT 2101 has 32 students with mean 71. Section 20 has 18 students with mean 79. A student says the overall mean is (71 + 79)/2 = 75. Is this correct? Find the overall mean.

B4 3-1 CLO2-A

The STAT 2101 outline weights are: In-Class Quizzes 15%, Lab Quiz 10%, Active Learning 5%, Midterm 25%, Final 45%. A student has 80, 90, 100 and 72 (out of 100) in the first four. (a) Can the student still reach an A (≥ 90)? (b) What final score secures A− (≥ 86)?

B5 3-1 CLO2-A · CLO3-B

Number of children in 40 families: 0 children → 5 families, 1 → 9, 2 → 14, 3 → 8, 4 → 4. Find the mean, median and mode, and describe the shape.

B6 3-1 CLO3-C

Choose the most appropriate measure of centre and justify it in one line each: (a) blood types of 200 donors; (b) customer satisfaction rated poor / fair / good / excellent; (c) house prices in Muscat, where a few villas cost over OMR 900,000; (d) heights of 500 adult women, roughly symmetric.

B7 3-1 CLO2-A · CLO3-C

Stem-and-leaf (key 3 | 2 = 32): 3 | 2 5 8 · 4 | 0 1 1 6 9 · 5 | 2 2 2 7 · 6 | 0 4 · 7 | 8. (a) Find the mean, median, mode and midrange. (b) If 78 had been recorded as 98, which of these four would change, and by how much?

B8 3-2 CLO2-A

For a sample, n = 12, $\Sigma X = 540$ and $\Sigma X^2 = 25{,}140$. Find the standard deviation and the coefficient of variation.

B9 3-2 CLO2-A

Find the mistake. For the *sample* 4, 7, 9, 12 a student wrote: $\bar{x} = 8$, $\Sigma(X - \bar{x})^2 = 34$, "$s^2 = 34/4 = 8.5$, $s = 2.92$". Identify the error and correct it.

B10 3-2 CLO2-A · CLO3-C

Over one summer in Muscat, the weekly petrol price had mean OMR 0.229 per litre with s = OMR 0.008, and the daily maximum temperature had mean 34 °C with s = 2.6 °C. Which variable was relatively more variable? Justify.

B11 3-2 CLO2-A

A data set has mean 80 and s = 3, and its shape is unknown. Find the interval, centred at the mean, that must contain **at least 96%** of the values.

B12 3-2 CLO2-A · CLO3-C

A data set has mean 50 and s = 5, and its shape is unknown. A student claims that at least 84% of the values lie between 40 and 65, because the upper limit is 3 SD above the mean. Evaluate the claim and give the correct guaranteed percentage.

B13 3-2 CLO2-A

Delivery times are bell-shaped, and about 95% of them fall between 18 and 42 minutes. (a) Find the mean and SD. (b) About what percentage take longer than 36 minutes? (c) About what percentage take between 24 and 42 minutes?

B14 3-2 CLO2-A · CLO3-C

Quiz marks out of 20 have mean 12.4, median 13, SD 3.1 and IQR 4.5. The instructor converts each mark to a score out of 100 (multiplying by 5) and then adds 4 bonus points. Find the new mean, median, SD, IQR and variance. Does the coefficient of variation change?

B15 3-2 CLO2-A

The deviations of five sample values from their mean are −3, 5, −1, d and 2. Find d and the sample variance.

B16 3-3 CLO2-A · CLO3-C

A scholarship requires a z-score of at least 1.75 on an entrance exam with mean 64 and s = 8. (a) What is the minimum qualifying score? (b) Hamad scored 75. Does he qualify?

B17 3-3 CLO3-C

In a 400 m race, Runner A ran 52 s in a group with mean 55 s and s = 2 s. Runner B ran 49 s in a group with mean 50 s and s = 0.8 s. Which runner performed better relative to their own group?

B18 3-3 CLO2-A

Daily visitors to a museum in Nizwa over 16 days (already ordered): 12, 15, 18, 20, 21, 23, 25, 26, 28, 30, 31, 33, 35, 38, 41, 47. (a) Find the percentile rank of 30. (b) Find P_{25} and P_{90}. (c) Find Q_1, Q_2, Q_3 by the median method and check for outliers. (d) Find D_3 and D_7.

B19 3-3 CLO2-A

In a class of 40 students, Maryam's percentile rank (by the course formula) is 81.25. How many students scored below her? How many scored above her, assuming no ties?

B20 3-3 CLO2-A · CLO3-C

Is 45 an outlier in the data 14, 18, 20, 22, 24, 26, 28, 30, 31, 45? Show the procedure.

B21 3-4 CLO3-B · CLO3-C

The five-number summary of waiting times (minutes) for 120 patients is 4, 11, 15, 22, 49. (a) Describe the shape. (b) Is 49 an outlier? (c) About how many patients waited at least 15 minutes? (d) What percentage waited between 11 and 49 minutes?

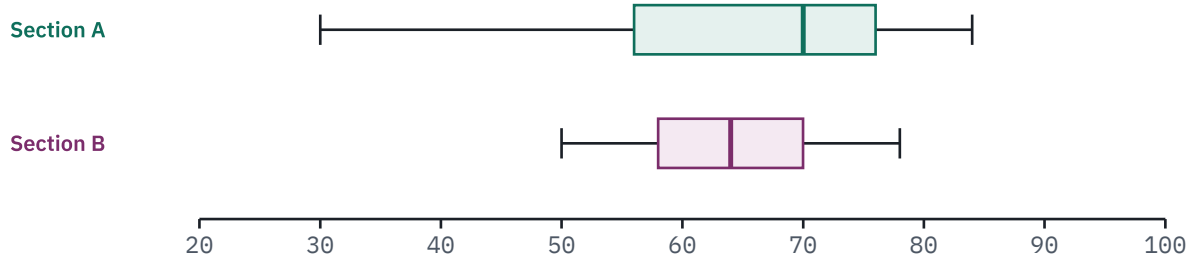
B22 3-4 CLO3-C

Five-number summaries of waiting times (minutes) at two bank branches: Muscat 3, 8, 12, 15, 21; Sohar 5, 7, 9, 18, 30. Compare the branches on centre, spread, range and shape.

B23 3-4 CLO2-A · CLO3-B

Using the stem-and-leaf data of B7, find the five-number summary, check for outliers, and describe the shape from the boxplot.

B24 3-4 CLO3-B · CLO3-C



Midterm scores (out of 100) for two STAT 2101 sections. Read values from the scale.

درجات الميدتيرم لشعبتين. اقرأ القيم من المحور.

(a) Read the five-number summary of each section. (b) Which section has the higher median? (c) Which is more variable in its middle 50%? (d) Describe each shape. (e) In which section did at least half the students score above 65?

Part C · Long structured questions

Midterm-style, multi-part, mixing lessons. Marks shown in brackets.

C1 3-1 3-2 3-3 3-4 CLO2-A · CLO3-B · CLO3-C **[18 marks]**

A health centre in Seeb recorded the number of patients on 12 randomly selected days: 23, 27, 18, 31, 25, 29, 35, 22, 27, 40, 26, 21.

1. Find the mean, median, mode and midrange.
2. Find the sample standard deviation using the shortcut formula, and the coefficient of variation.
3. Find Q_1, Q_2, Q_3 (median method), the IQR, and check for outliers.
4. Give the five-number summary and describe the shape of the boxplot. Is this consistent with your answer to part 1?
5. By Chebyshev's theorem, at least what percentage of days lie within $\bar{x} \pm 2s$? What percentage actually do?
6. Find the z-score of the busiest day and interpret it.

C2 3-1 3-2 CLO2-A · CLO3-C **[12 marks]**

Ages of a sample of 50 SQU staff members:

AGE	20–29	30–39	40–49	50–59	60–69
f	5	12	18	10	5

1. Find the mean, the sample standard deviation and the CV.
2. Find the modal class and the median class.
3. If the last class were "60 and over", which of your answers could still be found? Explain.

C3 3-2 CLO2-A · CLO3-C **[12 marks]**

Two factories fill 500 ml bottles. Factory A: mean 502 ml, s = 4 ml. Factory B: mean 498 ml, s = 2.5 ml.

1. Which factory is more consistent? Justify with a calculation.
2. Nothing is known about A's distribution. At least what percentage of A's bottles hold between 494 and 510 ml?
3. Out of 2,000 bottles from A, at most how many hold less than 490 or more than 514 ml?
4. B's fills are bell-shaped. About what percentage of B's bottles hold between 493 and 503 ml? About what percentage hold less than 495.5 ml?

C4 3-2 3-3 CLO2-A · CLO3-C **[12 marks]**

Midterm marks (out of 50) for 20 students, ordered: 18, 22, 25, 27, 29, 30, 31, 33, 34, 35, 36, 36, 38, 39, 40, 41, 43, 44, 46, 49. Here $\bar{x} = 34.8$ and $s = 8.121$.

1. Find the percentile rank of 35 and the value at P_{80}.
2. Khalid scored 41. Find his z-score.
3. The instructor converts each mark to a score out of 100 (× 2) and adds 3 bonus points. Find the new mean and SD.
4. Does Khalid's z-score change after the conversion? Does his percentile rank? Explain.

C5 3-1 3-2 3-3 3-4 CLO3-B · CLO3-C **[14 marks]**

Monthly electricity bills (OMR) for samples of 40 homes in two Muscat neighbourhoods:

	$\bar{X}$	MEDIAN	S	MIN	Q_1	Q_3	MAX
Al Khuwair	38.2	31.5	14.6	12	24	47	96
Al Amerat	29.4	29.0	6.1	15	25	34	44

1. Describe the shape of each distribution using two pieces of evidence.
2. Which neighbourhood's bills are more consistent? Justify.
3. Check each data set for outliers.
4. Which measure of centre should be reported for each neighbourhood? Why?
5. At least how many Al Amerat homes have bills between 17.2 and 41.6 OMR?

Part D · True / false with justification, and find the mistake

A one-word answer earns nothing. The justification is the mark.

D1 3-2 CLO1-A · CLO3-C

True or false? Justify. If the standard deviation of a data set is 0, all its values are equal.

D2 3-1 CLO1-A · CLO3-C

True or false? Justify. The median is always one of the data values.

D3 3-2 CLO1-A · CLO3-C

True or false? Justify. Adding the same constant to every value changes the coefficient of variation.

D4 3-2 CLO1-A · CLO3-C

True or false? Justify. Chebyshev's theorem says exactly 75% of values lie within 2 SD of the mean.

D5 3-4 CLO1-A · CLO3-C

True or false? Justify. If $Q_2 - Q_1 = Q_3 - Q_2$, the distribution must be symmetric.

D6 3-3 CLO1-A · CLO3-C

True or false? Justify. The IQR can never be larger than the range.

D7 3-1 CLO1-A · CLO3-C

True or false? Justify. In a negatively skewed distribution the mean is usually greater than the median.

D8 3-2 CLO1-A · CLO3-C

True or false? Justify. The variance of 5, 5, 5, 6 is larger than the variance of 1, 1, 1, 2.

D9 3-1 CLO2-A

Find the mistake. "The median of 9, 2, 7, 4, 6 is 7, the middle number."

D10 3-3 CLO2-A

Find the mistake. For 20 ordered values a student found P_{35} as follows: c = 20·35/100 = 7, "so P_{35} is the 7th value".

D11 3-3 CLO2-A

Find the mistake. For the data 10, 12, 15, 15, 18, 20, 25, 30 a student found the percentile rank of 15 as (4 + 0.5)/8 × 100 = 56.25.

D12 3-2 CLO3-C

Find the mistake. A student says data set B (mean 200, s = 20) is more variable than data set A (mean 10, s = 3) "because 20 > 3".

Part E · Dr. Ahmed's review self-study exercises

The three exercises that close Dr. Ahmed's deck, fully solved in the answers file.

E1 3-2 CLO2-A · CLO3-C

Review exercise (Dr. Ahmed). The average cost of a Talabat delivery is 26 OMR, and the **variance** of the cost is 9. One Thursday evening Talabat served only customers whose orders were **outside** the range 12 to 40 OMR. What is the maximum percentage of customers served that evening?

E2 3-1 3-2 3-3 CLO2-A **[14 marks]**

Review exercise (Dr. Ahmed). Weights (pounds) of 30 newborn babies: 5.4, 5.5, 5.7, 5.8, 5.9, 6.1, 6.1, 6.4, 6.4, 6.5, 6.6, 6.7, 6.7, 6.7, 7, 7, 7, 7.1, 7.2, 7.2, 7.4, 7.5, 7.7, 7.7, 7.8, 8, 8.1, 8.1, 8.3, 8.7.

1. Find the mode, median, mean and standard deviation.
2. Find the five-number summary and the interquartile range.
3. Find an interval centred at the mean that contains at least 64% of the data.
4. What percentage of the 30 weights actually lies in that interval?

E3 3-1 CLO3-B · CLO3-C

Review exercise (Dr. Ahmed). STAT 2101 students collected the annual maximum temperature (°C) in Muscat for ten years from 2011: mean = 51.3, mode = 46.5, median = 52. What is the shape of the distribution?

Part F · In-Class Quiz I simulation (20 marks · 30 minutes)

Closed book, calculator allowed. Time yourself before opening the answers file.

F1 3-1 CLO1-A

[3 marks] Parameter or statistic? (a) The mean GPA of all 2025 SQU graduates. (b) The SD of travel time for 60 students surveyed at the library. (c) The proportion of 500 sampled Omani households owning an electric car.

F2 3-1 CLO2-A · CLO3-C

[4 marks] The number of emails received on 8 days: 8, 12, 15, 15, 17, 21, 24, 40. Find the mean, median and mode. Which best describes a typical day? Why?

F3 3-2 CLO2-A

[4 marks] A sample of five values: 6, 9, 10, 13, 17. Using the shortcut formula, find the standard deviation and the coefficient of variation.

F4 3-2 CLO2-A

[3 marks] A sample of 80 commute times has mean 120 minutes and s = 15. Without assuming any shape, at least how many commutes lie between 82.5 and 157.5 minutes?

F5 3-3 CLO3-C

[3 marks] Aisha scored 72 in Statistics (class mean 64, s = 5) and 81 in Calculus (class mean 75, s = 4). In which subject is her relative position better?

F6 3-4 CLO3-B

[3 marks] The five-number summary of a data set is 20, 45, 70, 75, 80. Describe the shape and give two pieces of evidence. What percentage of values are at most 45?

Chapter 3 study set: 8 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's Spring 2024 slides, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief: what the outline tests · 01 · Lesson 3-1: Measures of Central Tendency · 02 · Lesson 3-2: Measures of Variation · 03 · Lesson 3-3: Measures of Position · 04 · Lesson 3-4: Exploratory Data Analysis · 05 · Revision pack: formulas, decisions, traps, R · **06 · Exam-style questions (no answers)** · 07 · Full worked answers